EXPERIMENT-08:

Write a program to implement Linear Regression (LR) algorithm in python

CODE:

```
# Import required libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# Create a sample dataset
X = np.array([[1], [2], [3], [4], [5],
              [6], [7], [8], [9], [10]])
y = np.array([2, 4, 5, 4, 5, 7, 8, 9, 10, 12])

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create the Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)
```

```
# Predict the test data
y_pred = model.predict(X_test)

# Display model results
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Plot the regression line
plt.scatter(X, y, color="blue", label="Actual Data")
plt.plot(X, model.predict(X), color="red", label="Regression Line")

plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear Regression")
plt.legend()
plt.show()
```

OUTPUT:

```
Slope: 1.0689655172413794
Intercept: 0.6206896551724128
Mean Squared Error: 0.7996432818073731
R2 Score: 0.9111507464658475
```

